
CHAPTER 20

Detection of T-Cell Degranulation: CD107a and b

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I. Introduction: Methods to Assess CD8⁺ T-Cell Function

T lymphocyte-mediated cytotoxicity has historically been examined using the ⁵¹Cr release assay, wherein antigen-expressing radioactively labeled target cells are recognized and killed by specific T cells (Brunner *et al.*, 1968). Though cumbersome, these assays directly assessed perhaps the most critical function of cytolytic T cells: the ability to kill target cells. Unfortunately, these assays tend to be nonquantitative and insensitive; and no information is gained about the number or more importantly the identity of the effector T cells.

Flow cytometric analysis of CD8⁺ T lymphocyte-mediated cytotoxicity has similar limitations, most importantly that the identity of cytolytic CD8⁺ T cells (the cells responsible for eliciting cytotoxic activity) is not directly examined. Several flow-based killing assays have been developed, which we and others have used to detect CD8⁺ T-cell-mediated cytotoxicity (Betts *et al.*, 2003; Fischer *et al.*, 2002; Lecoeur *et al.*, 2001; Liu *et al.*, 2002; Ritchie *et al.*, 2000; Sheehy *et al.*, 2001). One such assay system uses fluorescent dyes, such as carboxy fluorescein diacetate succinimidyl ester (CFSE), PKH-26, or chloromethyl-benzoyl-amino-tetramethyl-rhodamine (CMTMR) to label peptide-loaded target cells, which are then combined with CD8⁺ T cells (Betts *et al.*, 2003; Fischer *et al.*, 2002; Ritchie *et al.*, 2000; Sheehy *et al.*, 2001). As the peptide-specific CD8⁺ T cells kill the target cells, cytotoxicity is measured as the loss of fluorescently labeled target cells within the assay sample. This type of system can be combined with standard measures of apoptosis, such as 7-AAD or annexin V (Lecoeur *et al.*, 1997, 2001) staining to provide additional information about the target cells. Another method to assess cytotoxicity is the caspase assay (Chahroudi *et al.*, 2003; Liu *et al.*, 2002). This method detects the activation of caspase enzymes within target cells, an early process in the apoptotic pathways that is induced upon delivery of the “lethal hit” by cytolytic T cells. Unfortunately, however, none of these assays alone provides sufficient information regarding the identity, phenotype, or frequency of the CD8⁺ T cells that mediate the killing. Several methods have been developed to identify antigen-specific CD8⁺ T cells. These include, most notably, major histocompatibility complex (MHC) class I tetrameric complexes (Altman *et al.*, 1996) and intracellular cytokine staining (Betts *et al.*, 2000; Jung *et al.*, 1993; Kern *et al.*, 1998; Waldrop *et al.*, 1997). Although both of these techniques can provide valuable information regarding the frequency, phenotype, and/or the functionality of antigen-specific T cells, they do not examine the ability of the CD8⁺ T cells to elicit cytotoxic activity.

In this chapter, we describe an assay that serves to help bridge the gap between measurement of CD8⁺ T-cell cytolytic activity and the identity of the effector CD8⁺ T cells. To understand how this assay is performed, it is important to first review the fundamental mechanism by which CD8⁺ T cells mediate cytolytic activity—degranulation. Cytotoxic CD8⁺ T lymphocytes mediate the killing of target cells via two major pathways: perforin-granzyme-mediated activation of apoptosis and fas–fas ligand-mediated induction of apoptosis (Henkart, 1994; Nagata and Golstein, 1995; Sarin *et al.*, 1997; Trapani *et al.*, 1998). Induction of these pathways depends on the release of cytolytic granules from the responding CD8⁺ T cells (Bossi and Griffiths, 1999; Yannelli *et al.*, 1986). Degranulation is a necessary prerequisite to perforin-granzyme-mediated killing and is required for immediate lytic function mediated by responding antigen-specific CD8⁺ T cells. Cytotoxicity does not require *de novo* synthesis of proteins by the effector CD8⁺ T cell; instead, pre-formed lytic granules located within the cytoplasm are released in a polarized fashion toward the target cell (Barry and Bleackley, 2002; Griffiths, 1997). The lytic granules are membrane-bound secretory lysosomes that contain a

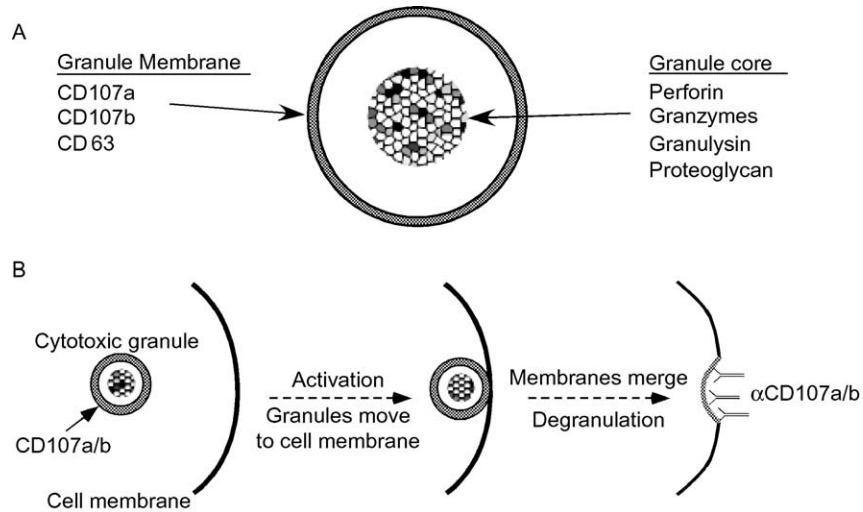


Fig. 1 Schematic representation of a CD8⁺ T-cell cytolytic granule (A) and the principle mechanism of degranulation as it relates to measurement by CD107a/b expression (B). (A) Cytotoxic granules are membrane-bound organelles related to lysosomes. A delimiting outer membrane contains various lysosomal membrane proteins, including CD107a, CD107b, and CD63. This surrounds a densely packed core, containing perforin, various granzymes, and granulysin, complexed with proteoglycan. (B) Upon activation of a CD8⁺ T-cell, pre-formed cytotoxic granules move toward the outer membrane of the cell in a polarized fashion toward the site of antigen presentation. Upon reaching the plasma membrane, the membrane of the granule and the plasma membrane merge, releasing the contents of the granule into the extracellular space. At this time, the CD107a and CD107b proteins from the granule membrane are accessible at the cell surface for specific antibody binding.

dense core composed of various proteins, including perforin and granzymes (Peters *et al.*, 1991) (Fig. 1A). The granule core is surrounded by a lipid bilayer containing numerous lysosomal-associated membrane glycoproteins (LAMPs), including CD107a (LAMP-1), CD107b (LAMP-2), and CD63 (LAMP-3).

During the process of degranulation, the lytic granule membrane merges with the plasma membrane of the activated CD8⁺ T cell (Peters *et al.*, 1991) and the contents of the granule are then released into the immunological synapse between the CD8⁺ T cell and the target cell (Fig. 1B). As a result of this process, the granular membrane, including CD107a, CD107b, and CD63 glycoproteins therein, is incorporated into the plasma membrane of the responding CD8⁺ T cell (Fukuda, 1991; Peters *et al.*, 1991). High-level expression of CD107a and b on the cell surface of activated T cells requires degranulation, because degranulation inhibitors, such as colchicine, dramatically reduce cell-surface expression of CD107a and b (Betts *et al.*, 2003). Protein synthesis inhibitors, such as cycloheximide, do not prevent expression, indicating that cell-surface CD107a/b expression on activated T cells occurs as a result of degranulation and not *de novo* synthesis of the CD107 proteins (A. Oxenius, 2003, unpublished observations).

Importantly, these proteins are rarely found on the surface of resting T lymphocytes (Betts *et al.*, 2003; Fukuda, 1991; Peters *et al.*, 1991). Thus, labeling responding cells with antibodies to CD107a and b and measuring their expression by flow cytometry can directly identify degranulating CD8⁺ T cells.

II. Materials and Methods

The measurement of CD107 expression on the cell surface of activated T cells can be used with numerous other measures of T lymphocyte function. We first present a general protocol for this technique and then present specific applications and pitfalls. The methods presented here all use human cells; however, these methods also apply to studies using murine cells, provided that the appropriate species-specific antibodies are used.

A. Materials

1. *Cells to be analyzed:* Fresh or frozen peripheral blood lymphocytes or T-cell clones can be used; viability is very important for background issues (see pitfalls), so samples with viability less than 85–90% will lead to spurious results.

2. *Antibody to CD107a and/or CD107b:* These antibodies are available from BD-Pharmingen (San Diego, California) in directly conjugated forms; the only commercially available direct conjugates include the following: for CD107a (clone H4A3), biotin, fluorescein isothiocyanate (FITC), phycoerythrin (PE), and Cy5-PE; and for CD107b (clone H4B4), FITC. The same antibodies are also available from Southern Biotech (Birmingham, Alabama) in purified form; these reagents must be titrated (see later discussion) for use in the assay. Note that CD107a is the higher expressed of the two molecules (Betts *et al.*, 2003) and, therefore, may itself be sufficient for degranulation analysis (Rubio *et al.*, 2003). Additional formats of CD107a and b, including allophycocyanin (APC) and Alexa 680 conjugates, were made using standard conjugation techniques (Roederer, 2003).

3. *Antibodies to CD28 and CD49d:* Purified azide-free antibodies to these molecules are added to provide costimulation, as is commonly used in intracellular cytokine assays; these antibodies are available from BD Biosciences (San Diego, California) and are used at a concentration of 1 µg/ml of each antibody.

4. Fluorescently labeled antibodies to various cell-surface markers (CD3, CD4, CD8, CD14, CD19) and cytokines (interferon-γ [IFN-γ], tumor necrosis factor-α [TNF-α], etc.).

5. *Appropriate antigen:* When examining peptide-specific CD8⁺ T-cell responses, use cytotoxic T lymphocyte (CTL) epitopes or peptide libraries at the desired peptide concentrations; *Staphylococcus enterotoxin B* (1 µg/ml final) serves as a positive control in all assays. Include unstimulated (or costimulation only) samples to control for background issues.

6. *Culture media*: Typically, RPMI-1640 supplemented with 10% fetal calf serum, antibiotics, and L-glutamine.

7. *Monensin*: Used to neutralize pH of endosomal and lysosomal compartments, and to prevent cytokine secretion; available from BD Biosciences (monensin [Golgistop], 0.7 μ l/ml final concentration); it is absolutely necessary to include this reagent in all stimulations (see [section IV](#)).

8. *Brefeldin A (BFA)*: Used as a secretion inhibitor; only necessary if TNF- α production is to be examined ([O'Neil-Andersen and Lawrence, 2002](#)); 10 mg/ml in dimethyl sulfoxide (DMSO) stock concentration, used at a final concentration of 10 μ g/ml.

9. *Wash buffer*: 1% bovine serum albumin and 0.1% sodium azide in phosphate-buffered saline (PBS), store at 4 °C.

10. *Fixative*: 1% paraformaldehyde in PBS, store at 4 °C.

11. *Permeabilization solution*: Available directly from BD Pharmingen, Cytofix/Cytoperm kit.

B. Initial Assay Setup: Titration of CD107 Antibodies

It is particularly important to initially determine the amount of CD107a/b antibodies to be included in the culture stimulation. Unlike typical antibody-staining procedures, the CD107a and b antibodies must be included throughout the duration of the stimulation period (see [section IV](#)).

1. Adjust cells to a concentration of 1×10^6 cells/ml in culture media. Transfer cells to a 5-ml polystyrene tube (Falcon no. 2058, Becton Dickinson), 1 ml/tube.

2. Add 1 μ g of SEB to each of six tubes containing 1×10^6 cells.

3. In parallel, prepare six tubes containing 1×10^6 cells with no antigen added.

4. Add 0.7 μ l of monensin to each tube.

5. Add in serial dilutions of CD107a or b antibody into each SEB-stimulated and unstimulated tube. If using CD107a and b antibody direct conjugates from BD Pharmingen, start at an initial concentration of 20 μ l/ml, followed by twofold serial dilutions.

6. Incubate for 5–6 hours at 37 °C in a 5% CO₂ incubator.

7. Wash cells with wash buffer, 5 minutes at 1500 rpm; re-suspend pellet.

8. (*Optional*): Permeabilize the cells using Cytofix/Cytoperm kit, per manufacturer's instructions. It is especially useful, though not necessary, to co-stain the activated cells with a cytokine marker, to ensure that measurement of CD107a/b expression is found on most cytokine-producing CD8⁺ T cells. Determination of the optimum amount of CD107a/b antibody to use is also simplified by having another positive response marker.

9. Stain cells with appropriate T-cell markers. It is particularly important to include anti-CD3, to help differentiate background CD107a/b staining. When examining SEB-specific responses, it is helpful to include an antibody to CD8, because the vast majority of T-cell-expressed CD107a/b will be found on this subset. If cells were permeabilized, stain also with anti-IFN- γ , to provide an internal positive control for the SEB response.

10. Wash cells with wash buffer, 5 minutes at 1800 rpm; re-suspend pellet in 1% PFA.

11. Analyze by flow cytometry. Data analysis and gating strategy is shown in Fig. 2. Determine the minimum amount of CD107a/b antibody necessary to provide the best separation and population size of the responding cells while keeping background staining to a minimum.

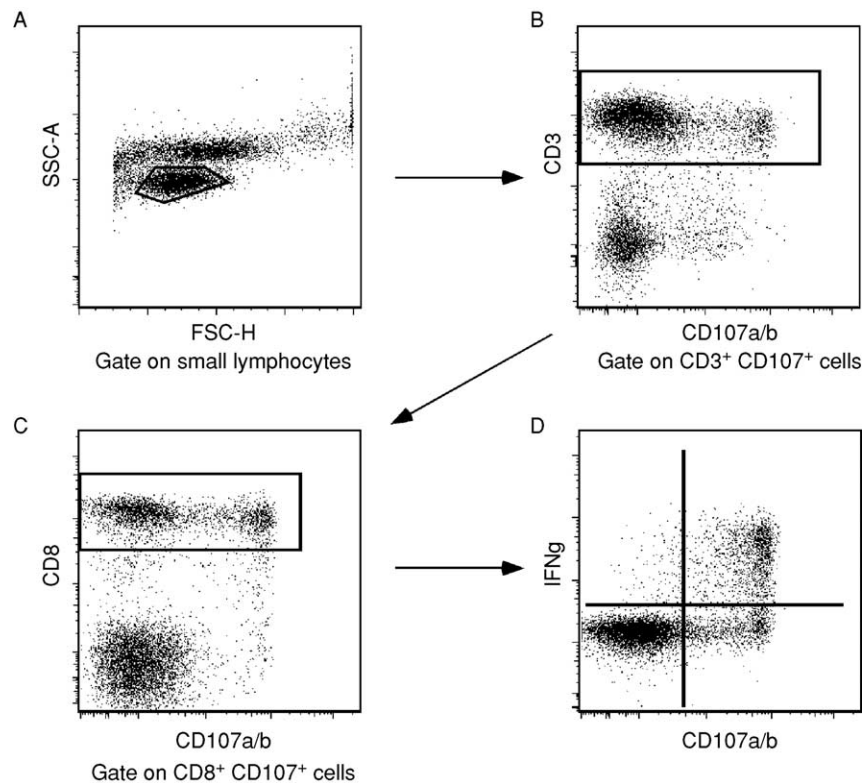


Fig. 2 Gating schematic for analysis of CD107a and b expression on SEB-stimulated cells. The initial gate is placed around small resting lymphocytes on a forward scatter channel versus a side scatter channel plot (A). Next, the gated cells are subgated based on CD3 expression to analyze on T cells (B). The CD3⁺ T cells are then further subdivided into CD8⁺ and CD8⁻ cells, and CD8⁺ events are gated upon (C). Finally, CD107a/b expression is plotted versus IFN- γ secretion using a quadrant analysis to differentiate various subpopulations of cells based on response type (D).

C. Antigen-Specific CD8⁺ T-Cell Degranulation

Once the optimum amount of CD107a/b antibody to include in the assay is determined, antigen-specific CD8⁺ T-cell responses can be examined. It becomes especially important, and more interesting, to combine CD107a/b measurement and intracellular cytokine staining together, thus allowing the examination of separate CD8⁺ T-cell functions simultaneously. The general assay setup is nearly identical to the methods used when titrating the CD107a and b antibodies.

1. Prepare a master cell mix, containing the following:

Cells diluted to 1×10^6 cells/ml in culture media

0.7 μ l/ml monensin

1 μ l/ml each of anti-CD28 and anti-CD49d antibodies

1 μ l/ml BFA, if TNF- α production is to be examined

Appropriate amount of CD107a/b antibodies, as determined by titration

2. Aliquot master cell mix into a 5-ml polystyrene tube, 1 ml/tube.
3. Add the following antigens and controls to each tube containing 1×10^6 cells:
No antigen (negative control)
SEB, as previously (positive control)

Peptide(s)/antigen of interest: The concentration of peptide should be determined previously or a peptide titration curve can be used to determine optimal concentrations. Typically, 1–2 μ M peptide concentration should give maximal responses, although this can vary. Note that if peptides are diluted in DMSO, the final concentration of DMSO in the culture volume cannot exceed 1% without a substantial negative impact on cell response and viability.

4. Incubate for 5–6 hours at 37 °C in a 5% CO₂ incubator.
5. Wash cells with wash buffer, 5 minutes at 1500 rpm; re-suspend pellet in minimal volume.
6. Stain cells with any antibodies requiring surface staining.
7. Permeabilize the cells using Cytofix/Cytoperm kit, per manufacturer's instructions.
8. Wash cells with Cytofix wash buffer twice, 5 minutes at 1800 rpm; re-suspend pellet in minimal volume.
9. Stain cells with cytokine antibodies. Note that it is preferable to stain cells with anti-CD3 and anti-CD8 antibodies after they are permeabilized. This serves to partially, if not fully, eliminate the reduced staining of these molecules on the cell surface of activated T cells due to surface downmodulation of these markers after activation.

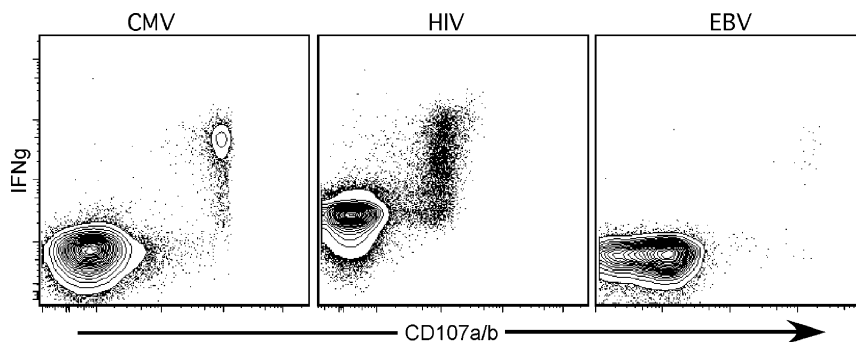


Fig. 3 Representative examples of antigen-specific expression of CD107a/b on CD8⁺ T cells after activation with viral peptides. PBMCs from three patients were stimulated with peptides derived from cytomegalovirus (left), human immunodeficiency virus (middle), or Epstein–Barr virus (right). After incubation, cells were analyzed as described in Fig. 2. Note the presence of CD107a/b-positive IFN-positive cells in each panel, as well as the different types and level of the responses observed.

10. Wash cells with wash buffer, 5 minutes at 1800 rpm; re-suspend pellet in 1% PFA.

11. Analyze by flow cytometry. Three examples of antigen-specific responses wherein the responding CD8⁺ T cells both degranulate and produce cytokine upon activation with viral peptides derived from human cytomegalovirus (CMV), human immunodeficiency virus (HIV), or Epstein–Barr virus (EBV) are shown in Fig. 3.

III. Results

An example of the gating strategy to analyze CD107a/b expression in activated CD8⁺ T cells is shown in Fig. 2. Cells were stimulated with SEB, as described in section II, and collected on a FACSCalibur flow cytometer, using FITC, PE, PerCP, and APC as the fluorescent parameters. The initial gate is on small resting lymphocytes (Fig. 2A), with care taken to avoid events with low side scatter and low forward scatter, as well as events with high side scatter (granulocytes/macrophages). This is particularly important for reducing background staining. Next, CD3⁺ events are gated upon (Fig. 2B). At this stage, one must account for any downregulation of CD3 that may have occurred in the activated T cells, if the CD3 stain was performed on the cell surface rather than intracellularly. If downregulation has occurred and is not accounted for, a significant proportion of the responding cells may be inadvertently gated out. In the example shown, the CD3 stain was performed on permeabilized cells, thus little downregulation of CD3 from the surface of the activated cells is observed. Also notice in this panel that the T cells expressing CD107a/b are very bright, as compared to the low-level expression observed in the CD3-negative cells. After the CD3⁺ T-cell gate, the cells are now separated into CD8⁺ and CD8[−] events (Fig. 2C). As was the case for the previous

CD3 gate, care must again be taken to account for any cells that have down-regulated CD8 in the process of becoming activated. In this example, the CD8 stain was again performed on permeabilized cells. It can also be seen here that a small but detectable portion of $CD3^+/CD8^-$ cells also becomes CD107a/b positive in response to SEB; these likely represent degranulating $CD4^+$ T cells, and a similar type of analysis can be performed on this T-cell subset. Finally, if the cells were co-stained with a cytokine marker, such as IFN- γ , the relationship between cytokine production and degranulation can be directly observed by plotting IFN- γ versus CD107a/b expression in the $CD8^+$ (Fig. 2D) or $CD4^+$ T cells. As can be seen, although most SEB-responding $CD8^+$ T cells become CD107a/b positive, the expression of IFN- γ can be quite variable and a substantial proportion of responding cells do not produce IFN- γ . This serves in part to illustrate the diversity of the $CD8^+$ T-cell response, because not all the responding cells exhibit the same functional profile.

Figure 3 shows three examples of responses obtained using specific peptides derived either from CMV, HIV, or EBV and illustrates some of the different response profiles that may be observed and the level of sensitivity that can be achieved when analyzing very small responses. In all three examples, human PBMCs were stimulated with antigenic peptides using the method described in section II. Figure 3 (left panel) shows the response to a CMV peptide in a CMV-positive individual. As can be seen, all the cells producing IFN- γ also become CD107a/b positive. The center panel shows the response to a mixture of overlapping 15-mer peptides derived from HIV Pol in an HIV-positive individual. Unlike the CMV response in the left panel, a significant proportion of responding cells to the HIV peptides do not produce IFN- γ , and a varying degree of IFN- γ is observed in those cells that do produce this cytokine. Both the CMV- and the HIV-specific responses shown here are relatively large responses, on the order of 2–7% of $CD8^+$ T cells. In the right panel, however, a much smaller yet detectable response to an EBV-derived peptide is shown. Although this response is very low, approximately 0.2% of $CD8^+$ T cells, it is still easily distinguishable as a predominantly IFN- γ^+ CD107a/b $^+$ response.

IV. Critical Aspects

As with other techniques to measure T-cell function by flow cytometry, proper data analysis is a crucial aspect of this technique. Unlike standard intracellular cytokine assays, there are specific issues with examination of CD107a/b expression that must be considered during both the assay setup and the data analysis.

A. Trafficking of CD107a and b

CD107a and b in $CD8^+$ T cells are localized to lysosomal, endosomal, and cytotoxic granular membranes (Peters *et al.*, 1991). This localization occurs as a result of a sorting signal within the cytoplasmic tail of these proteins (Chang *et al.*,

2002; Gough and Fambrough, 1997; Guarnieri *et al.*, 1993; Honing and Hunziker, 1995). Furthermore, cell-surface expression of these proteins on CD8⁺ T cells is typically very low and is retrieved from the cell surface via an endocytic pathway into an endosomal or lysosomal compartment (Chang *et al.*, 2002; Fukuda, 1991). Because of this active internalization of the CD107a and b molecules, it is necessary to label CD8⁺ T cells with antibodies to these molecules during the activation period (Betts *et al.*, 2003). Failure to include antibodies during the stimulation can have a dramatic effect on the ability to stain cell-surface CD107a and b. As shown in Fig. 4, cells labeled throughout the stimulation period have a significantly enhanced staining pattern compared to cells stained at the end of the stimulation period.

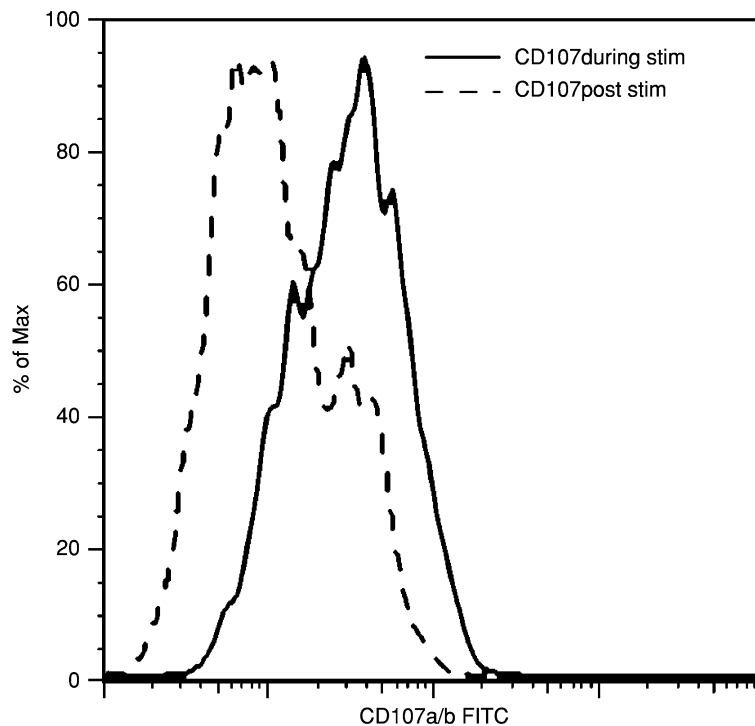


Fig. 4 Histogram analysis of CD107a/b expression on SEB-stimulated cells stained either during the stimulation period (*solid line*) or after the stimulation period (*dashed line*). Analysis of the cells was performed as described in Fig. 2. Gates were placed on CD3⁺/CD8⁺ cells shown to respond to SEB stimulation by producing IFN- γ . Histograms of CD107a/b levels on the IFN- γ -producing cells were then compared, demonstrating that cells stained with CD107a/b after activation have substantially lower levels of surface CD107a/b.

B. Monensin Requirement

The active trafficking of CD107a and b from the cell surface introduces yet another wrinkle into the staining procedure. When the antibody-fluorochrome-bound CD107a or b molecules are reinternalized, they will traffic into endosomal or lysosomal compartments, where they may be exposed to a highly acidic environment and various proteases. This environment has a dramatic effect on the fluorescence of various fluorochromes, because of either acid sensitivity or protease sensitivity. To reduce this effect, one should include monensin throughout the stimulation period. Monensin serves to neutralize the pH of the endosomal and lysosomal compartments (Mollenhauer *et al.*, 1990), reducing the loss of fluorescence of various fluorochromes exposed to this environment. As shown in Fig. 5, this effect is not limited only to small organic fluorochromes sensitive to pH, such as FITC and Alexa 680, but also on proteinaceous fluorochromes such as PE and APC (Roederer *et al.*, 1987). The latter effect likely occurs because of altered trafficking of proteases within the endosomal and lysosomal compartments as a result of monensin-induced deacidification of these compartments (Chapman and Munro, 1994). As can be seen in Fig. 5, stimulation in the absence of monensin leads to the substantial loss of the fluorescent signal obtained from FITC, PE, APC, and Alexa 680-conjugated anti CD107a/b in activated cells. The inclusion of monensin also serves to prevent the secretion of various cytokines, such as IFN- γ , MIP1 β , and IL-2 (Jung *et al.*, 1993; Lee *et al.*, 1990). It must be noted, however, that monensin does not adequately prevent the secretion of TNF- α from activated cells (O'Neil-Andersen and Lawrence, 2002). Thus, if CD107a/b expression and TNF- α production are to be monitored in CD8⁺ T cells, both monensin and BFA must be used in combination.

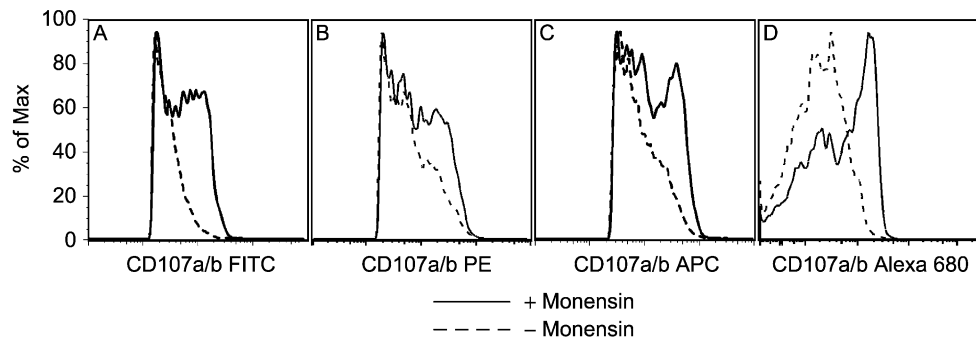


Fig. 5 Monensin is necessary to obtain optimal staining of CD107a/b conjugated with fluorescein isothiocyanate (A), phycoerythrin (B), allophycocyanin (C), or Alexa 680 (D). Cells were stimulated with SEB in the presence of CD107a/b antibodies, with (solid line) or without (dashed line) monensin. Histograms represent the level of CD107a/b expression on CD3⁺/CD8⁺ cells that responded to SEB by IFN- γ production.

C. Background Expression of CD107a and b

Background expression of CD107a and b on the surface of T cells is typically very low in peripheral blood, although they can be detected on the cell surface of mitogen-activated T cells (Kannan *et al.*, 1996). It should be noted, however, that these proteins are not expressed only in T lymphocytes. These proteins are normal residents of endosomal and lysosomal membranes and are detectable at a relatively high copy number in all cell types in humans, mice, rats, and chickens (Fukuda, 1991). Some peripheral blood cells, including activated platelets and monocytes, express these molecules on the cell surface to a higher degree than other cell types (Eskelinen *et al.*, 2003; Silverstein and Febbraio, 1992). Additionally, because all cell types express these proteins, cells with a permeable membrane such as dead or dying cells will stain positive for CD107a and/or b. Thus, it is extremely important for the cells to be tested to have high viability to help reduce background concerns. An additional concern regarding background arises as a result of the CD107a and b staining procedures. By including the antibodies to these proteins during the stimulation period, antigen-presenting cells present within the cells being examined, specifically B cells and monocytes, readily uptake the antibody for processing. Because monensin is present, these cells are unable to proteolytically process the antibody/fluorochromes and, therefore, also stain positive for CD107a and b. This effect can be readily seen in the example shown in Fig. 6 (upper left panel). Monocytes and B cells, labeled here with CD14 or CD19, are all CD107a/b^{dull}. By gating on small lymphocytes (Fig. 6, upper right panel), the majority of the CD107a/b^{dull} cells are removed. By removing the remaining CD14/19-positive events within the small lymphocyte gate (Fig. 6, bottom panels), a substantial proportion of the remaining CD107a/b background staining is eliminated, leaving the antigen-specific T-cell-expressed CD107a and b intact for further analysis. Thus, whenever possible, inclusion of a dump channel, specifically to label monocytes, B cells, and dead cells (if possible), can significantly reduce the CD107a and b background. Despite these measures, occasional patient samples will have higher CD107 background than others, with expression levels varying from 0.01% to more than 0.5% of CD8⁺ T cells. Once the background has been reduced by the means presented here, the remaining background likely represents either nonspecific or an unknown antigen-specific degranulation by the CD8⁺ T cells. Typically, however, the expression levels tend to be quite dull. In these cases, it may be pertinent to use a stricter gating method only on CD107a/b^{bright} events.

===== V. Applications

The general applications for this technique revolve primarily around the examination of antigen-specific T lymphocyte function in humans and mice. This is the first flow cytometry-based assay that directly assesses the ability of T lymphocytes to degranulate. At the most basic level, the functionality of the *ex vivo* CD8⁺ T-cell response to specific individual peptides of viral origin can be examined in a

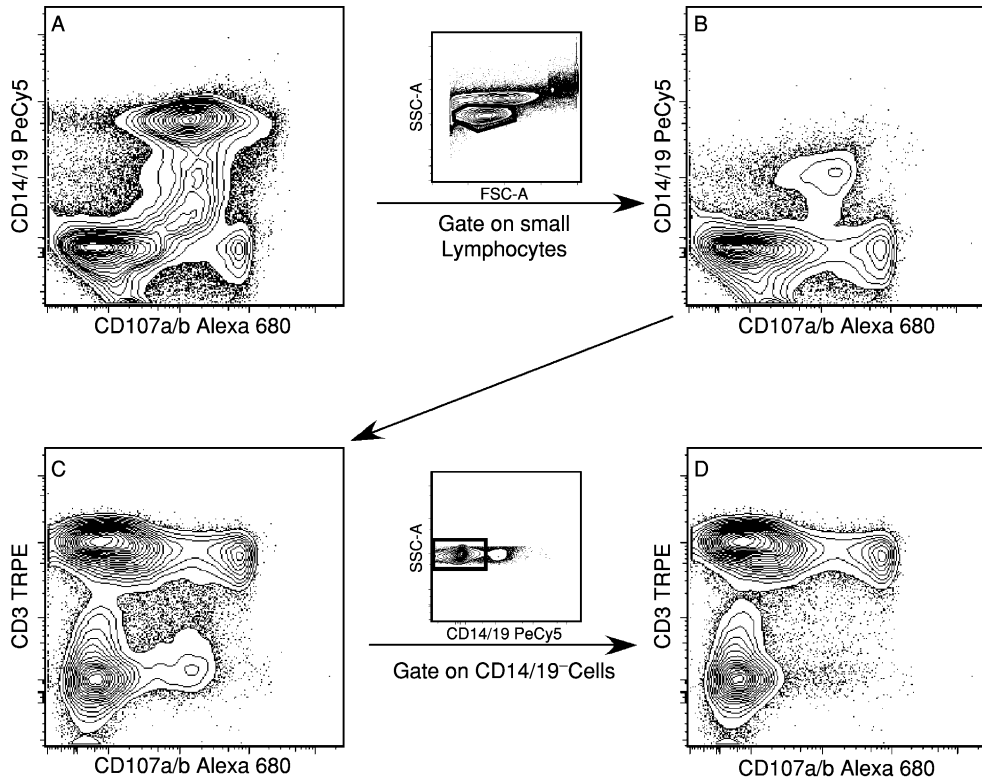


Fig. 6 Removal of background CD107a/b expression through gating and CD14/CD19 staining. (A) Ungated SEB-stimulated cells are shown to demonstrate the level of CD107a/b labeling found on CD14/CD19⁺ cells. Note that the level of CD107a/b found on the CD14/CD19⁺ events is intermediate, as compared to the CD107a/b bright cells that do not co-stain with CD14/CD19. (B) By gating on small resting lymphocytes, a large proportion of CD107a/b dull events are removed. Note that this gating did not affect the CD107a/b bright events. Also, a small proportion of CD14/CD19⁺ cells remain within the small lymphocyte gate and are CD107a/b dull. (C) Changing the parameters compared in (B), we can see that a large proportion of CD107a/b dull events do not co-stain with the T-cell marker CD3. (D) Removing the remaining CD14/CD19⁺ events by negative gating dramatically reduces the remaining CD107a/b background expression on the non-T cells.

more complete manner by measuring cytokine production and degranulation simultaneously. With polychromatic flow cytometry, multiple cytokines and degranulation can be measured simultaneously in activated CD4⁺ and CD8⁺ T lymphocytes.

Aside from simply providing another quantitative measure of functionality for T cells, measurement of degranulation allows a direct link between cytolytic potential and cytokine production. Measurement of degranulation provides a positive marker for cytolytic activity of effector CD8⁺ T cells, unlike measurement of perforin in CD8⁺ T cells, which is lost if the cells degranulate as a result of

activation (Appay *et al.*, 2000). There is a direct correlation between the ability of human CD8⁺ T-cell clones to kill, as measured by ⁵¹Cr release assay, and the level of cell-surface CD107a expression after stimulation (Rubio *et al.*, 2003).

CD107a/b expression is also highly amenable for use in live cell sorting of activated antigen-specific T cells, because no fixation or permeabilization of the cells is necessary. Current methods to sort live antigen-specific CD8⁺ T cells relies on either MHC class I tetramer, which is highly restricted by reagent availability, or cytokine-secretion capture systems, which tend to be of low sensitivity. Cell sorting using CD107a/b expression is highly sensitive and is not limited by reagent availability, because peptides or whole antigens are used, obviating the need for patient- or MHC-specific reagents.

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